

Dear staff of the International Journal of Public Health,

We have updated the files to meet the journal requirements. A comment has been added to the author proof document for each of the 21 queries.

All figures have been replaced with higher-resolution versions. For Figure 2B, a formatting error was corrected where correlations were incorrectly displayed as percentages. We applied the same correction to Figures S1–S4 in the supplementary material and we also slightly adapted the caption. A corrected version has been uploaded.

Citation issues (queries Q10, Q16–Q18) can be addressed in the following way:

- Q10: The citation 33 should be cited at the end of the following sentence: "While extreme weather or pollution could still contribute, previous studies have suggested that they only explain a minor fraction of the observed excess mortality."
- Q16: The citation 20 was changed to include the published article instead of the preprint. Here is the correct citation:
G. Riutort-Mayol, P.-C. Bürkner, M. R. Andersen, A. Solin, and A. Vehtari, "Practical Hilbert space approximate Bayesian Gaussian processes for probabilistic programming," *Statistics and Computing*, vol. 33, no. 1, p. 17, Dec. 2022, doi: [10.1007/s11222-022-10167-2](https://doi.org/10.1007/s11222-022-10167-2).
- Q16: The citation was adapted to include the correct information. Here is the correct citation 32:
A. Carollo, H. Putter, P. H. Eilers, and J. Gampe, "Competing risks models with two time scales," *Statistical Methods in Medical Research*, vol. 34, no. 11, p. 9622802251367443, Sep. 2025, doi: [10.1177/09622802251367443](https://doi.org/10.1177/09622802251367443).
- Q17: The pages are 1–33. Here is the correct citation 25:
C. L. Hansen, C. Viboud, and L. Simonsen, "Disentangling the relationship between cancer mortality and COVID-19 in the US," *eLife*, vol. 13, pp. 1–33, Aug. 2024, doi: [10.7554/eLife.93758](https://doi.org/10.7554/eLife.93758).
- Q18: The citation 25 was adapted with a working DOI. Here is the correct citation:
C. L. Hansen, C. Viboud, and L. Simonsen, "Disentangling the relationship between cancer mortality and COVID-19 in the US," *eLife*, vol. 13, pp. 1–33, Aug. 2024, doi: [10.7554/eLife.93758](https://doi.org/10.7554/eLife.93758).

For query Q13, funding information needs to be adapted as:

"This work was supported by the Swiss Federal Office of Public Health (mandate 142006323)".

In addition, in the main manuscript, the placeholder "XX" in the GitHub URL, used for anonymity during review, needs to be replaced with the correct address. This should read:

"All code is available from <https://github.com/anthonyhauser/causes-of-deaths>."

For query Q21, the Alt-text has also been revised and is provided below.

"Figure 1 Alt-Text – Panel A is a bar chart showing the distribution of deaths by age class, with percentages increasing from less than 1% in ages 0 to 17 to 62% in ages 80 and over. Panel B is a line graph depicting the distribution of deaths over causes for different age classes, with cancers and cardiovascular causes dominant in older groups and a spike in the "Other" category for ages 0 to 17.

Figure 2 Alt-Text – Panel A shows time series line graphs for six causes of death in people aged 80 and older from 2011 to 2021, with observed data points in red, black trend lines, and a vertical dashed line marking the pandemic onset. Panel B presents a correlation matrix heatmap for different causes of death, using a brown color scale to indicate strength of correlation. Panel C is a dot plot revealing the timing and relative peak size of mortality by cause, differentiating dots before and during the pandemic, with dot color indicating period and dot size representing relative peak size.

Figure 3 Alt-Text – Panel A presents a series of line charts by age group (0–17, 18–39, 40–64, 65–79, 80+) showing expected deaths versus observed deaths from 2011 to 2021, with an increase in observed deaths, especially after the dashed line marking early 2020. Panel B shows excess mortality line charts for the same age groups from January 2020 to December 2021, comparing all-cause excess mortality including and excluding COVID-19, with higher excess deaths seen mostly in older age groups when accounting for COVID-19.

Figure 4 Alt-Text – Twelve-panel figure shows relative excess by phase and cumulative excess for cardiovascular, cancers, respiratory, and mental/neurological deaths by age group: 18–39, 40–64, 65–79, and 80+, from January 2020 to December 2021. Each panel uses colored circles with error bars to indicate COVID phases and excess mortality, with a black line tracing cumulative excess. Patterns of excess vary by condition, age group, and pandemic phase, illustrated by different circle colors and positions along the y-axes.

Figure 5 Alt-Text – Line chart with six panels compares partial correlation across lag weeks for COVID-19 and respiratory diseases as causes of death in individuals aged 65–79 and 80+. Panels are organized by cardiovascular, cancer, and mental or neurological causes. Each line, red for COVID-19 and blue for respiratory diseases, shows trends with shaded credibility intervals. Vertical dashed lines indicate the lag with the highest absolute partial correlation. Data highlight differing correlation patterns by cause and age group.”

Please let us know if any further adjustments are needed.

Sincerely,

Anthony Hauser